

Austin Norby, Ph.D.

Bogart Associates

Director of Internal Research and Development

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Current Position

Sept 2023 – Present: Director of Internal Research and Development, Bogart Associates

- Managed a novel research effort within Bogart Associates to combine machine learning and vulnerability research to predict the likelihood of vulnerable functions in IoT devices and firmware (ongoing)
- Managed and applied the Agile Methodology to refactoring a legacy product owned by Bogart Associates. Identifying the main components of the remaining code and determining where the real functionality exists. Developing a plan to devote resources to recreate missing functionality including UI, network protocols, Perfect Forward Encryption (PFE), and storage of encrypted data at rest as well as in transit.
- Developed a custom, 40-hour, internal training for technical people on cyber security engineering including basic capability development on both Windows and Linux operating systems, reverse engineering PE and ELF files using both static and dynamic analysis, vulnerability research, and culminating in a hands-on exercise to develop their own basic capability and demonstrate its effectiveness to the rest of the class.

May 2022 – Present: Contractor Team Lead, Bogart Associates

- Improved the vulnerability research team by increasing the number of members on the team from one to five, implementing agile methodologies (including sprint planning and retrospectives), and delivering a total of 8 releases of five capabilities in 2.5 years up from one release of one capability in five years; a 16x increase in productivity for the team.
- Lead and manage team of 5 vulnerability researchers, capability development specialists, and software engineers to find zero-day exploits and develop proofs-of-concept into cyber capabilities.
- Develop and improve developer experience by incorporating industry best practice code analysis, static analysis, linting, unit testing, integration testing, and CI/CD practices into developer standard operating procedure during capability development.
- Mentored hundreds of junior employees since 2019 through technical challenges and Agile methodology to include: reverse engineering, C, C++, Assembly, Python, Sprint Planning, Retrospectives, measuring team velocity, which lead to an overall positive culture of continuous learning and professional development
- Static and dynamic reverse engineering using Ghidra, WinDBG, IDA Pro, and custom tools as to fix complex bugs in user- and kernel-mode programs.

- Developed cyber capabilities using Visual Studio, VS Code, and VIM in C, C++, and Assembly languages.
 - Research, development, and reverse engineering experience related to modern anti-virus evasion techniques.
 - Research, development, and reverse engineering experience related to modern exploit development techniques.
 - Used Python to enhance personal workflow and improve testing accuracy and reproducibility for capabilities.
 - Worked on projects using many different build systems including Make, CMake, MSBuild, and custom, proprietary build systems.
 - Engineered a Proof-of-Concept code sample into a full capability including additional features, testing, and a scriptable, repeatable release processes.
 - Extensive use and understanding of IP networking including capability development, scripting, testing, networking management, DNS resolution, and tunneling protocols.
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Major Accomplishments

Dissertation

Norby, Austin, "A Cross-Architecture Framework for Anti-Debugging Techniques" (2024), Masters Theses & Doctoral Dissertations. 448. (<https://scholar.dsu.edu/theses/448>)

Conference Speaker

- SecTor, 2024, Toronto, CA
 - NopEmulator
 - This tool emulates instructions using the Ghidra Framework in order to determine if the bytecode is a Nop sled building on previous work in academia and improving the detection heuristic through actual emulation and removing length restrictions and focusing on the Intel x64 architecture. The algorithms and findings that support this specific implementation within Ghidra can be extended to support reverse engineering, malware analysis, and network-based exploit detection.
- HackRedCon, 2024, Louisville, KY
 - No I Didn't: Exceeding Authorized Access and the CFAA
 - Would you know if you violated the Computer Fraud and Abuse Act? This important piece of legislation often appears when prosecuting computer-related crimes and is important to understand when conducting any type of red team engagements. In this talk, we will look at a very modern case that brought questions about "Exceeding Authorized Access" to the U.S. Supreme Court. We'll quickly take a look at the details of the case, and ultimately what the impacts of the Supreme Court decision are on modern tech landscape.

Past Positions

Oct 2017 – May 2022: Software Engineer, Blue Star Software

- Static and dynamic reverse engineering using Ghidra, WinDBG, IDA Pro, and custom tools to fix complex bugs in user- and kernel-mode programs.
- Developed cyber capabilities using Visual Studio, VS Code, and VIM in C, C++, and Assembly languages.
- Research, development, and reverse engineering experience related to modern anti-virus evasion techniques.
- Used Python to enhance personal workflow and improve testing accuracy and reproducibility for capabilities.
- Worked on projects using many different build systems including Make, CMake, MSBuild, and custom, proprietary build systems.
- Engineered a Proof-of-Concept code sample into a full capability including additional features, testing, and a scriptable, repeatable release processes.
- Extensive use and understanding of IP networking including capability development, scripting, testing, networking management, DNS resolution, and tunneling protocols.
- Java Development, Docker, Networking, UI Development
- Quickly learned new technologies and applied directly to team's workflow to increase productivity
- Created UI demonstrations for future-proofing team's main product and to provide a clean, modern interface for customer testing and use
- Enterprise Java Development using a multi-stage build process
- Provided timely solutions based on customer needs and requirements
- Team environment, collaborated during development, testing, and review processes
- Used Atlassian suite of tools for Agile Development, SVN for version control
- System administration duties to set up, tear down, manage, configure, and network virtual machines using vSphere as necessary for development purposes.

Feb 2014 – Oct 2017: Cryptologic Computer Scientist

- Interfaced directly with management to affect modern changes and improvements for the entire Development Program including additional rewards for those who went above and beyond
- Collaborated with other elite members within the Development Program to oversee day-to-day operations, plan monthly meetings, coordinate activities during and outside of working hours, and deal with individual disputes as they arose.
- Managed two committees that focused on social events to improve morale between employees and tours/briefings that focused on learning about all aspects of the Intelligence Community.
- Repeatedly referred to as driven, ambitious, confident, and willing to speak openly and honestly about a range of topics

Apr 2017 – Oct 2017: Embedded Devices Researcher with Network Research

- Conducted highly technical, bleeding edge research on embedded devices looking for vulnerabilities and misconfigurations

- Worked successfully with a small, elite team to create a proof of concept for future operational use
- Sophisticated routing, switching, and networking knowledge including programming technologies, IP, TCP/UDP, HTTP, and media streaming were necessary to complete and present proof of concept to management
- Team was able to automate steps within proof of concept which provided an excellent presentation and seamless demonstration of capabilities

Sept 2016 – Apr 2017: Idea Champion in the CNOx Incubation Cell

- Selected by a Senior Government Official to research, architect, and develop an innovative idea to improve government Computer Network Operations
- Facilitated meetings, networking events, and interviews to acquire information to make informed decisions and prove the necessity of my innovative idea
- Acted as program manager, oversaw 10+ contractors to help develop my proof of concept including developers, testers, and technical writers
- Presented ideas and proof of concept to senior government officials; they selected the idea to be implemented immediately within a new CNO organization
- Presented CNOx Incubation cell process to organizations within the Department of Defense to improve marketing posture and attract individuals with unique ideas

Mar 2016 – Sept 2016: Capabilities Development Specialist

- Collaborated with team focused on the Windows Operating system and low level, Windows internals programming

Mar 2015 – Mar 2016: Exploitation Analyst

- Perform at a highly technical level as a mission practitioner, operator, analyst and researcher in order to learn and apply the theoretical foundations of Computer Science to mission problems and challenges
- Analyze cyber adversary tactics, techniques, and procedures (TTPs) to anticipate, identify, and counter the exploitation and/or network attack threats posed by those elements
- Identify network components and configurations, protocols, encryption devices, security measures, and analyzed packets in order to exploit target networks
- Utilize hex editors, debuggers, and other system internal tools to triage malware and other potentially harmful binaries, as well as investigate forensic artifacts
- Independently conducted technical research to facilitate target development and CNE activities

Sept 2014 – Mar 2015: Lead Data Scientist and Developer for the Office of Technical Counterintelligence

- Selected by the Office of Technical Counterintelligence management to spearhead their data science effort
- Self-starter, driven, and autonomous: Implemented the first proof-of-concept Apache Pig script to leverage available data feeds which laid the foundation for every future rotational employee, analyst and investigator associated with this project

- Developed a programmatic solution to enable non-technical analysts and investigators to perform technical queries on relevant data sources

Feb 2014 – Sept 2014: Lead Project Developer with Cryptanalytic Enterprise Solutions Division

- Lead Project developer; created a secure, web-based data repository allowing user to query and retrieve sensitive information as needed
- Individually, this project stored over 500k records in its first year and was used by hundreds of employees

May 2013 – August 2013: Cryptologic Access Program Intern within Access Operations Division

- Developed code to automatically perform daily temporal analysis in support of the detection of anomalous network activity using Python and R

April 2012 – May 2013: Software Engineer with Maverick Software Consulting

- Collaborated with Maverick interns and full-time, Thomson Reuters employees on Quality Assurance Team
- Learned and applied industry-standard techniques and software to test the speed and functionality of APIs designed for proprietary use by Thomson Reuters
- Developed proprietary framework to identify project baselines and improvements over time for web applications

June 2011 – April 2012: Undergraduate Teacher's Assistant with University of Minnesota – Twin Cities

- Discrete Structures of Mathematics Course
- Reinforced core concepts during lab sections
- Graded and tutored peers

Degrees, Certifications and Licenses

Dakota State University 2020 – 2024:

- Doctorate of Philosophy, Cyber Operations

Naval Postgraduate School 2015 – 2017:

- Masters of Engineering, Electrical Engineering

University of Minnesota – Twin Cities 2011 – 2013:

- Bachelor of Science in Computer Science
- Bachelor of Science in Mathematics

Certifications:

- (ISC)2 CISSP
- Offensive Security Certified Professional (OSCP)

- GIAC Certified Incident Handler (GCIH) (expired)
- GIAC Certified Forensic Analyst (GCFA) (expired)
- EC Council Certified Ethical Hacker (CEH) (expired)
- CompTIA Network+ (expired)

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